

TrES-5b

R-1993

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_150617 · created 2026-08-07T15:16:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457190.8252 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.121 +/- 0.046
Transit depth (Rp/Rs)^2	1.47 +/- 1.11 [%]
Orbital Inclination (inc)	82.46 +/- 2.41
Airmass coefficient 1 (a1)	0.894 +/- 0.023
Airmass coefficient 2 (a2)	0.088 +/- 0.022
Scatter in the residuals of the lightcurve fit is	2.61 %
Best Comparison Star	#3 - [325, 53]
Optimal Method	PSF photometry
Transit Duration (day)	0.052 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.121 ± 0.046 vs 0.142 (deviation 0.34σ)
Duration fit vs published	0.052 d vs 0.0758 d (deviation 0.71σ)
Mid-time O–C	4.5 min
Depth SNR	2.0
Residual scatter	2.61 %

Light curve

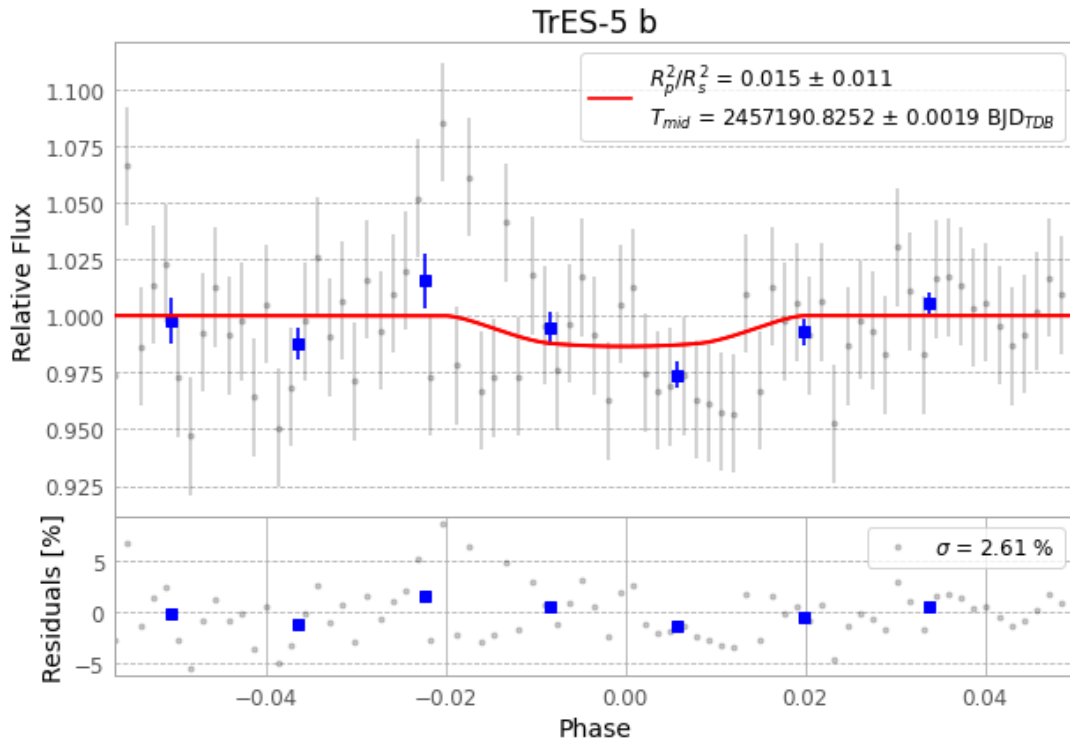


Figure 1 — FinalLightCurve_TrES-5 b_2015-06-16.png (SHA-256 3263a92ce80dc431...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [124, 251], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 0351d6ff4cb3a142...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), TRES-5150617054512.FITS → TRES-5150617093312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-150617.log (2e1956568620...), FinalLightCurve_TrES-5 b_2015-06-16.png (3263a92ce80d...), FinalLightCurve_TrES-5 b_2015-06-16.csv (3e2f442ed2d3...)

Compute resources

Wall time 126.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:16Z.